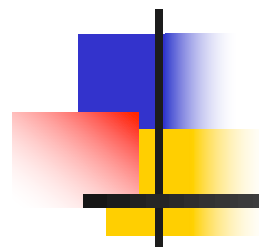
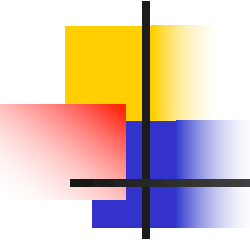


Building regression models

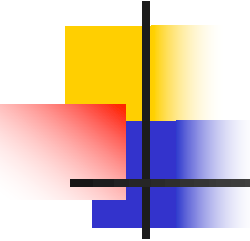


Univariate linear models



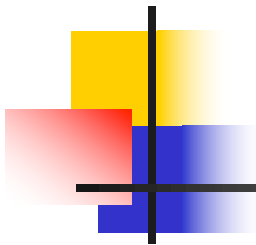
Regression and Correlation

- **Regression analysis** is *the process of constructing a mathematical model or function that can be used to predict or determine one variable by another variable.*
- **Correlation** is *a measure of the degree of relatedness of two variables.*



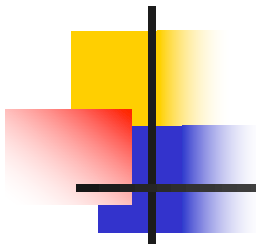
Simple Regression Analysis

- bivariate (two variables) linear regression -- the most elementary regression model
 - dependent variable, *the variable to be predicted*, usually called Y
 - independent variable, *the predictor or explanatory variable*, usually called X



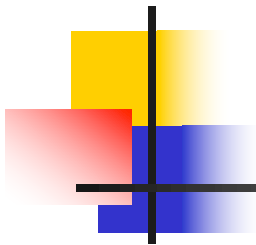
Scatter Plots

- In simple correlation and regression studies, data are collected on two quantitative variables (bivariate data) to determine whether a relationship exists between the two variables.
- To illustrate this graphically, consider the following example.



Scatter Plots

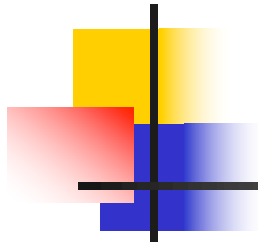
- **Example:** The bivariate data given in the following table relate the high temperature ($^{\circ}\text{F}$) reached on a given day and the number of cans of soft drinks sold from a particular vending machine in front of a grocery store. Data were collected for 15 different days.



Scatter Plots

Temp.	Quantity	Temp.	Quantity	Temp.	Quantity
70	30	98	59	90	53
75	31	72	33	95	56
80	40	75	38	98	62
90	52	75	32	91	51
93	57	80	45	98	58

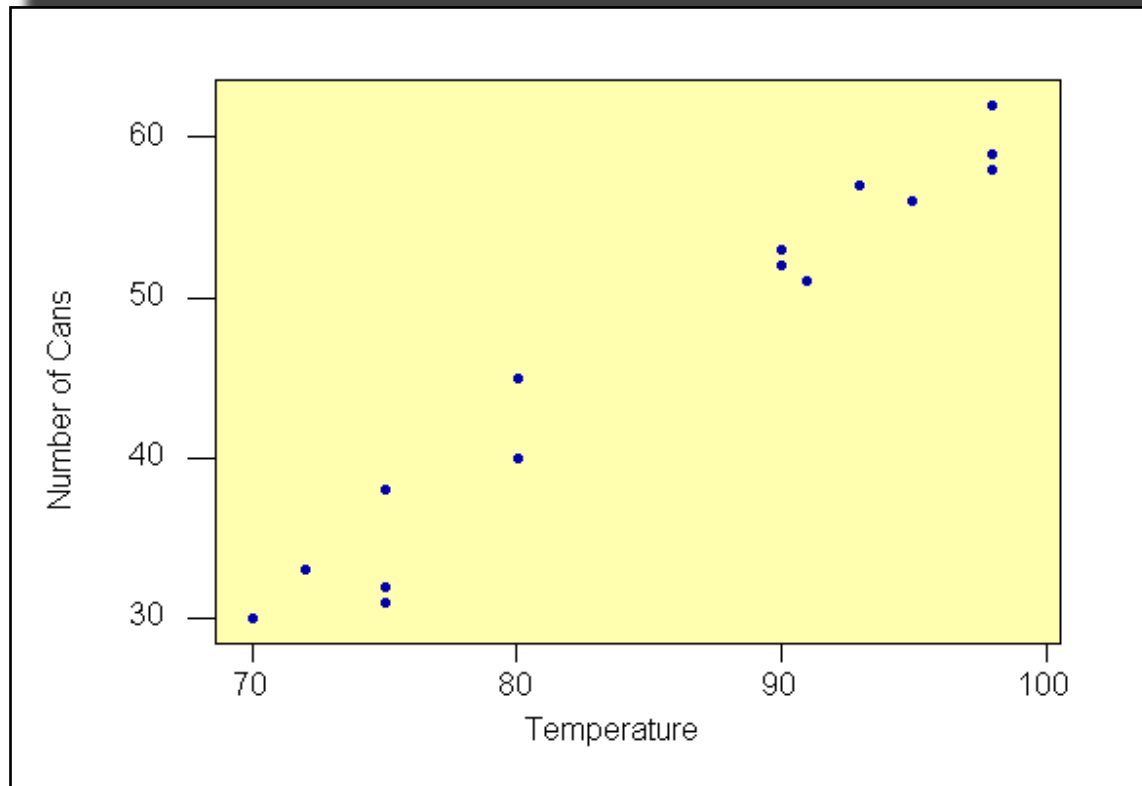
We would like to graphically study the association between the temperature and the number of cans of soft drinks sold.



Scatter Plots

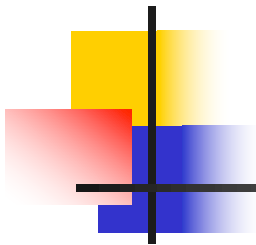
- To analyze graphically, we can display the data on a two-dimensional graph.
- We can plot the number of cans of soft drinks along the vertical axis and the temperature along the horizontal axis.
- Such plots are called **scatter plots**.

Scatter Plots



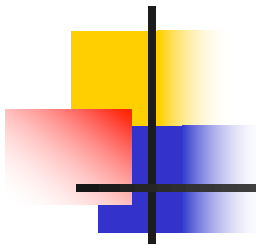
Observe that the number of cans of soft drinks sold increases as the temperature increases.

Scatter Plot of Number of Cans
versus Temperature



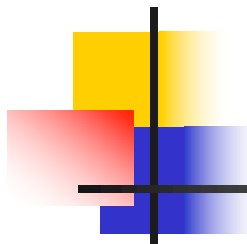
Scatter Plots

- The variable plotted along the vertical axis is called the **dependent variable**.
- The variable plotted along the horizontal axis is called the **independent variable**.
- Notation: We will let **y** represent the dependent variable and we will let **x** represent the independent variable.



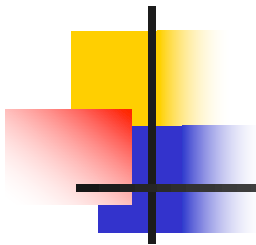
Scatter Plots

- **NOTE:** The number of cans of soft drinks sold will depend on the temperature.
- Thus, the dependent variable (y) will be the number of cans of soft drinks sold, and the independent variable (x) will be the temperature.



Looking at patterns

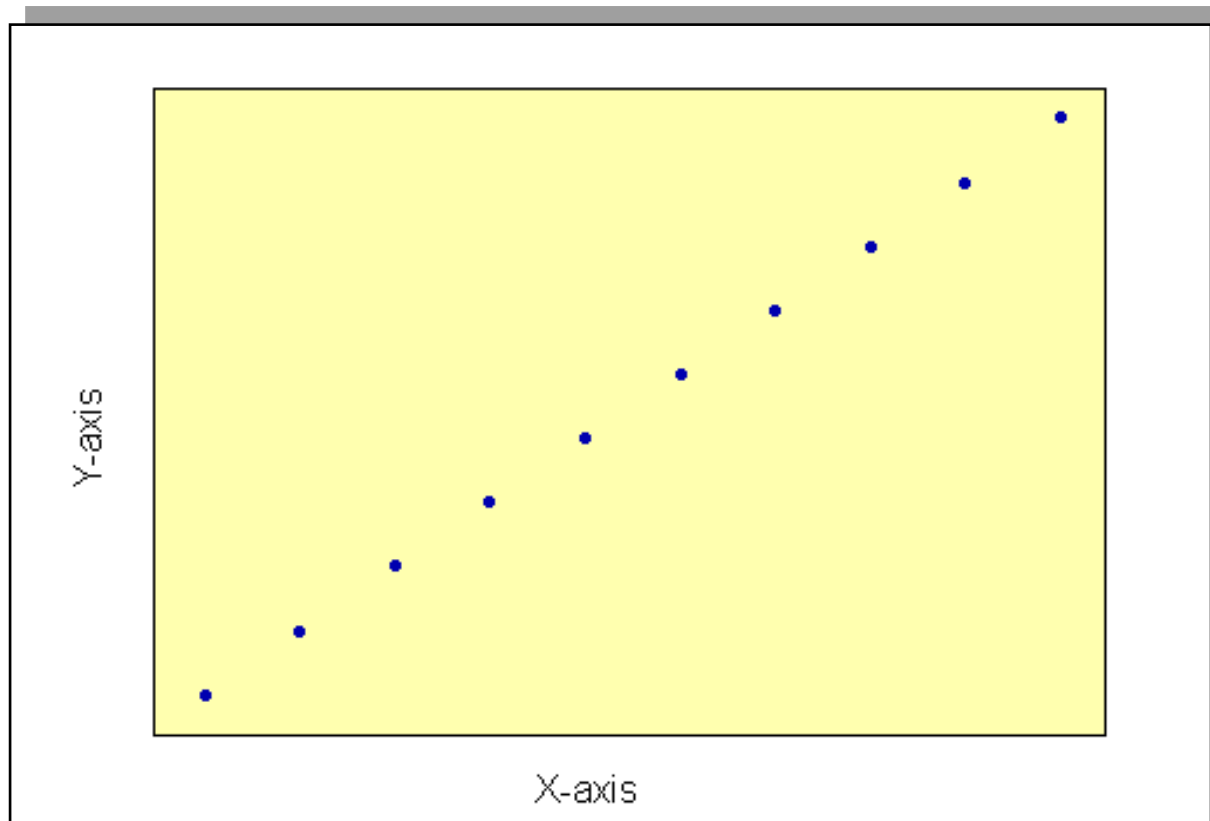
- Detecting an association or a relationship for bivariate data starts with a scatter plot.
- When examining a scatter plot, one should try to answer the following questions:
 - Is there a straight-line pattern or association?
 - Does the pattern or association slope upward or downward?
 - Are the plotted values tightly clustered together in the pattern or widely separated?
 - Are there noticeable deviations from the pattern?



Quick Tips

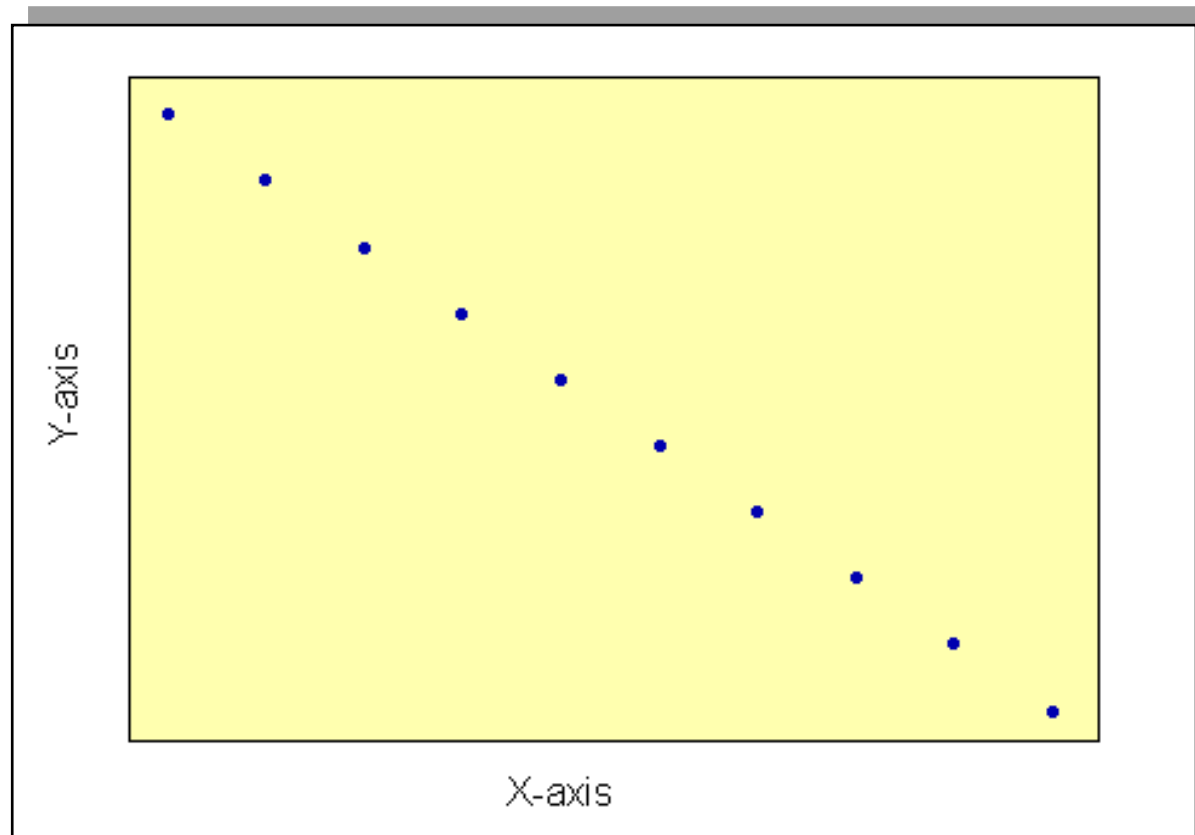
- Two variables are said to be **positively related** if larger values of one variable tend to be associated with larger values of the other.
- Two variables are said to be **negatively related** if larger values of one variable tend to be associated with smaller values of the other.

Perfect Positive Linear Association



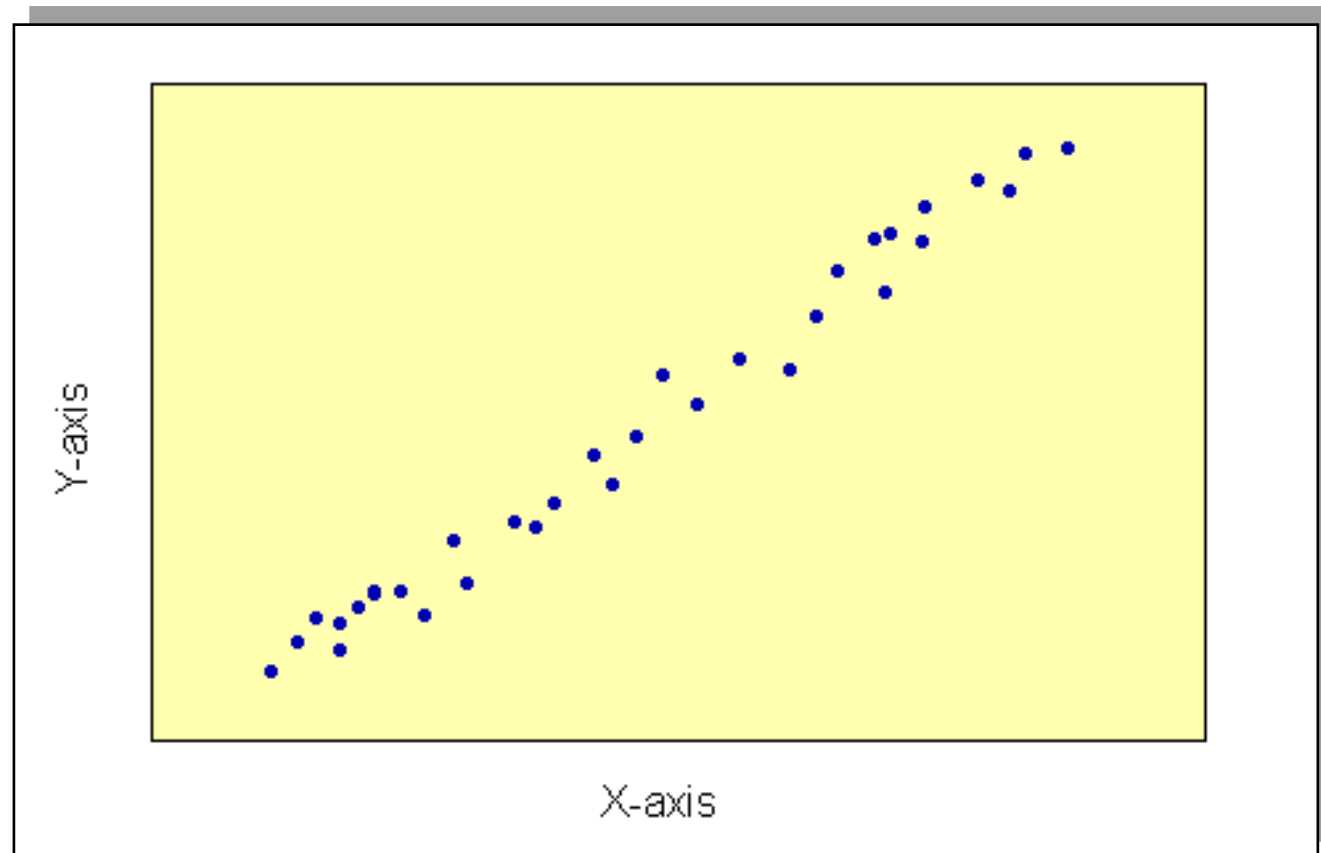
Perfect positive association rarely occurs
when sample data are collected.

Perfect Negative Linear Association



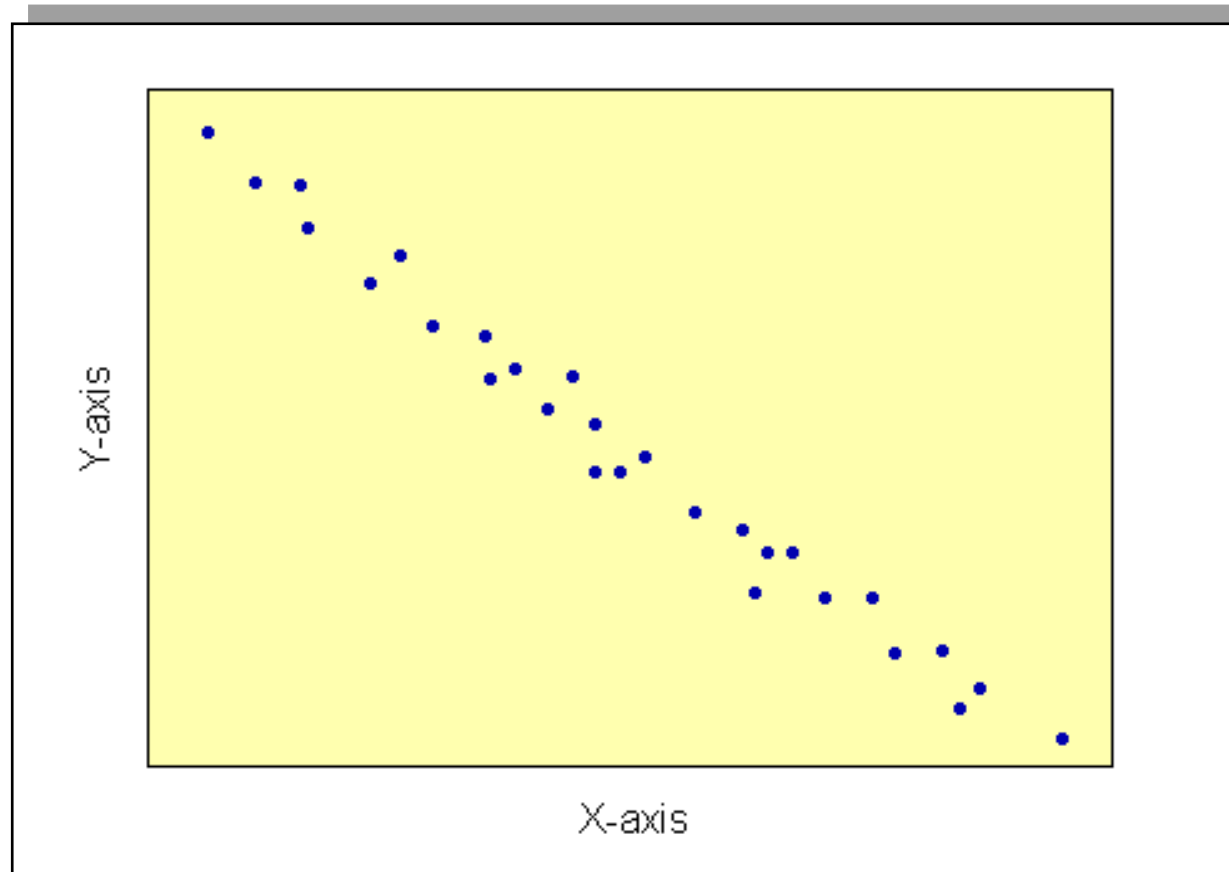
Perfect negative association rarely occurs
when sample data are collected.

Very Strong Positive Linear Association



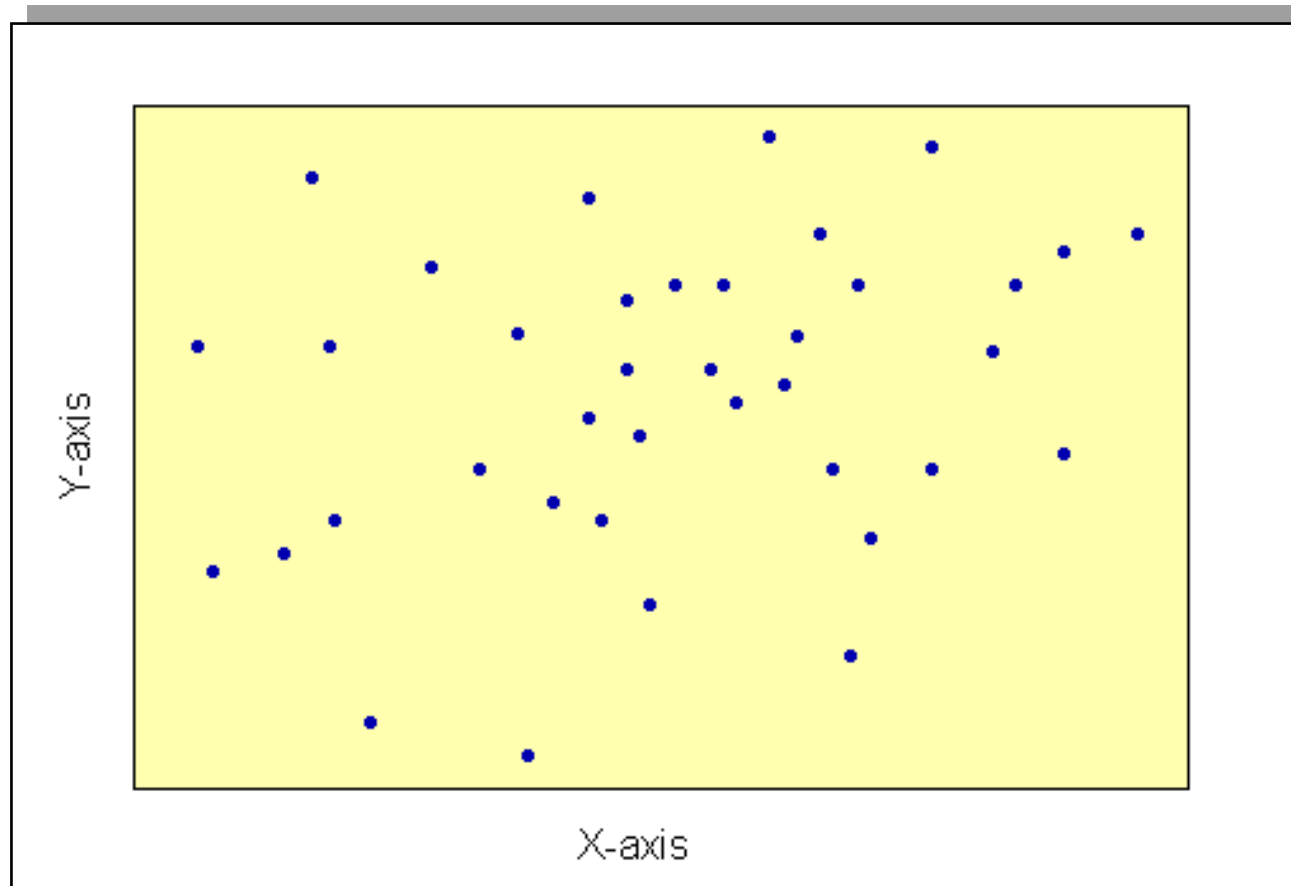
The points are closely packed along a positive linear trend..

Very Strong Negative Linear Association



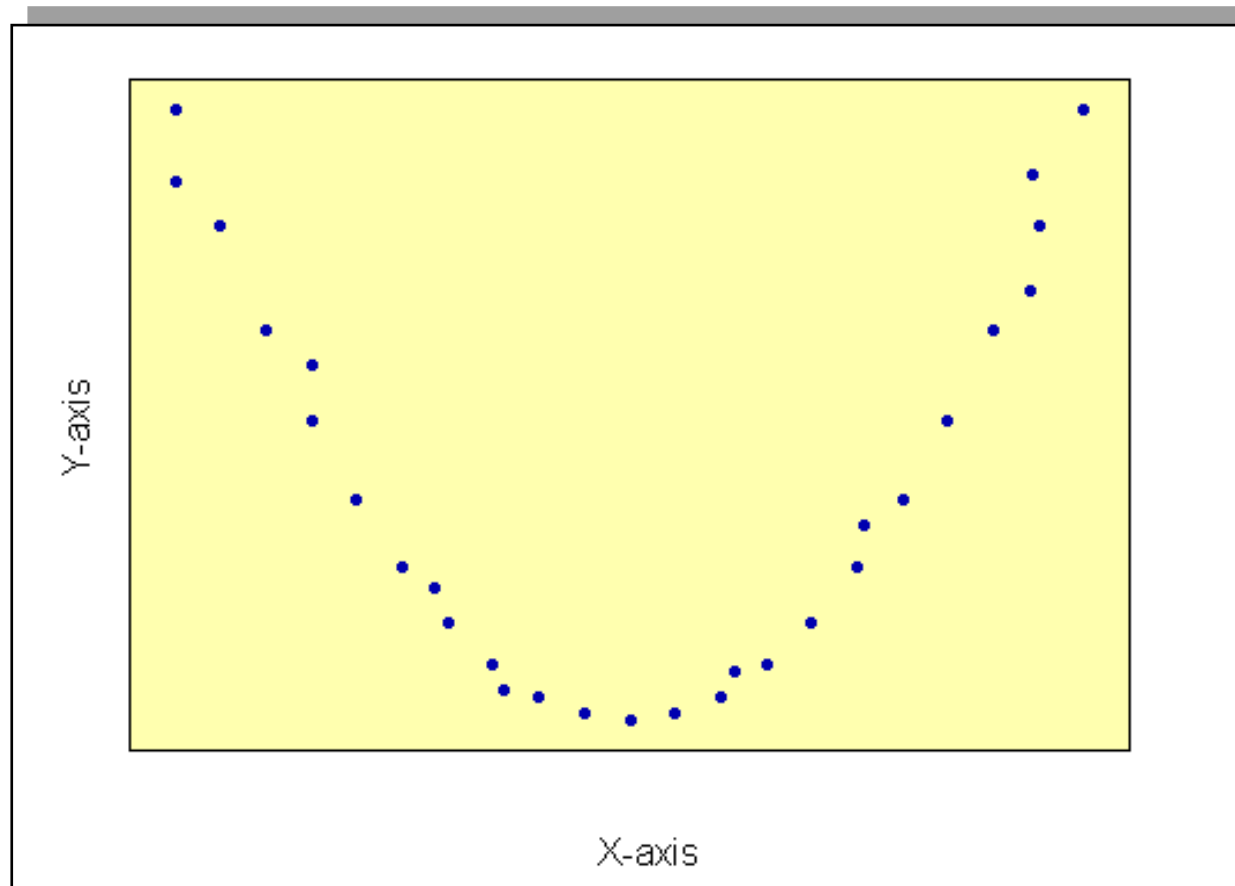
The points are closely packed along a negative linear trend..

Little or No Association

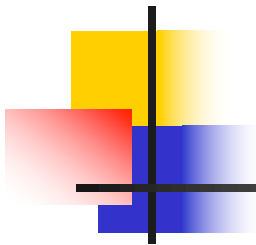


The points are scattered around with no apparent trend..

Nonlinear Association

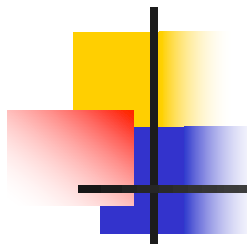


The points display a nonlinear relationship..



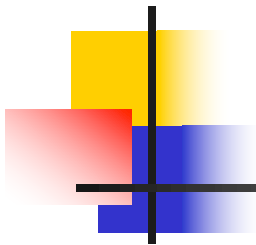
Correlation

- So far you have seen how a scatter plot can provide a visual of the association between two variables.
- Here we will discuss a numerical measure of the linear association between two variables called the **Pearson product moment correlation coefficient** or simply the **correlation coefficient**.



Correlation

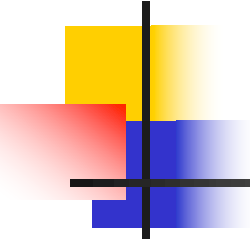
- Explanation of the term – sample correlation coefficient: The sample correlation coefficient measures the **strength** and **direction** of the **linear relationship** between two variables using sample data.
- The sample correlation coefficient is denoted by the letter ***r*** and is computed from the equation on the next slide.



Correlation

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{\{n(\sum x^2) - (\sum x)^2\} \times \{n(\sum y^2) - (\sum y)^2\}}}$$

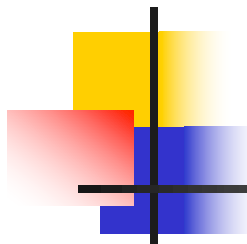
n is the number of (x,y) data pairs.



Correlation

- **Example:** Compute the linear correlation coefficient for the following set of observations for the independent variable x and the dependent variable y .

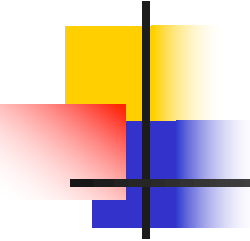
x	8	4	5	-1
y	-2	0	2	6



Correlation

- **Solution:** The formula may look intimidating, but we can construct a table to help with the computations.

x	y	$x \cdot y$	x^2	y^2
8	-2	-16	64	4
4	0	0	16	0
5	2	10	25	4
-1	6	-6	1	36
$\sum x = 16$	$\sum y = 6$	$\sum x \cdot y = -12$	$\sum x^2 = 106$	$\sum y^2 = 44$



Correlation

- **Solution:** Using the values from the previous table, we have

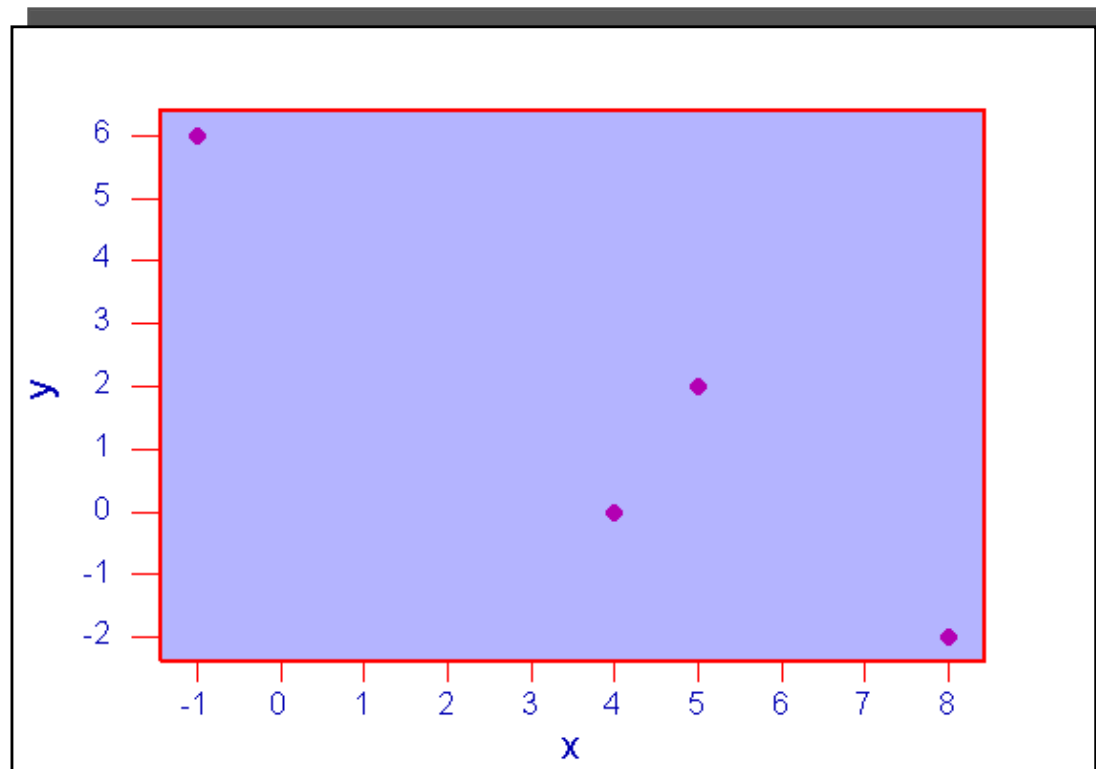
$$r = \frac{4(-12) - (16)(6)}{\sqrt{\{4(106) - (16)^2\} \times \{4(44) - (6)^2\}}} = -0.939$$

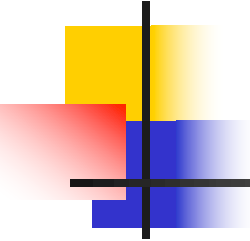
R:

```
x<-c(8,4,5,-1)
y<-c(-2,0,2,6)
a<-array(c(x,y),c(4,2))
r<-cor(a)
```

Correlation

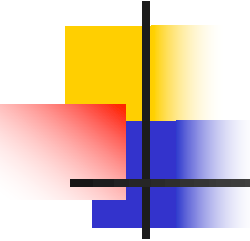
- The scatter plot displays the negative correlation between x and y .





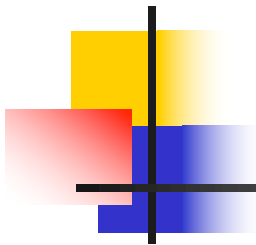
Properties of the Correlation Coefficient

- The range of the correlation coefficient is from -1 to $+1$.
- If there is a perfect positive linear relationship between the variables, the value of r will be equal to $+1$.
- If there is a perfect negative linear relationship between the variables, the value of r will be equal to -1 .



Properties of the Correlation Coefficient

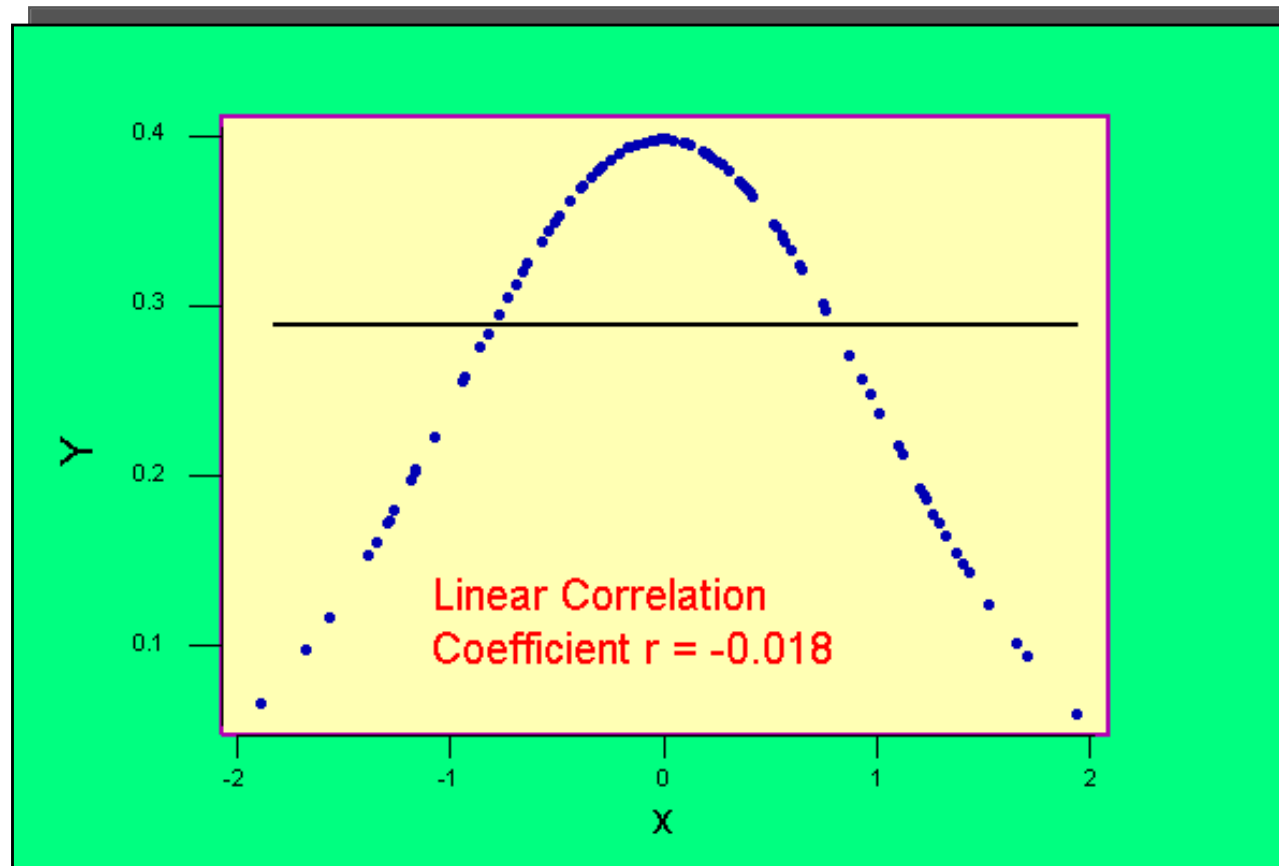
- If there is a strong positive linear relationship between the variables, the value of r will be close to $+1$
- If there is a strong negative linear relationship between the variables, the value of r will be close to -1
- If there is little or no linear relationship between the variables, the value of r will be close to 0 .



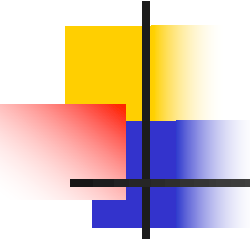
Quick Tip

- One should always examine the scatter plot and not just rely on the value of the linear correlation.
- This measure will not detect curvilinear or other types of complex relationships.
- That is, there may be a non-linear relationship between two variables even though the linear correlation is close to 0.

Quick Tip

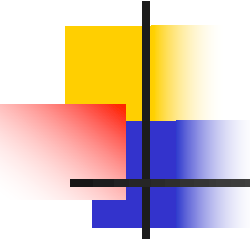


Small linear correlation but strong non-linear correlation.



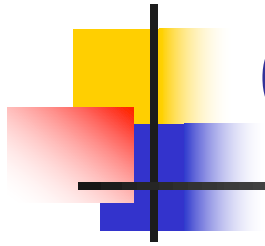
Correlation and Causation

- Suppose that a high correlation is observed between the weekly sales of hot chocolate and the number of skiing accidents.
- One can reasonably conclude that hot chocolate sales could not cause skiers to have accidents while skiing.
- Also, one can reasonably conclude that more skiing accidents could not cause an increase in hot chocolate sales.



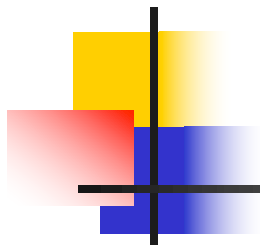
Correlation and Causation

- Since the two variables are not actually related, what could explain such a relationship?
- The apparent relationship between the two variables may be caused by a third variable.
- In this case, the variables may be related to the weather conditions during the winter months.



Correlation and Causation

**Hence, one can conclude
that correlation is not the
same as causation.**

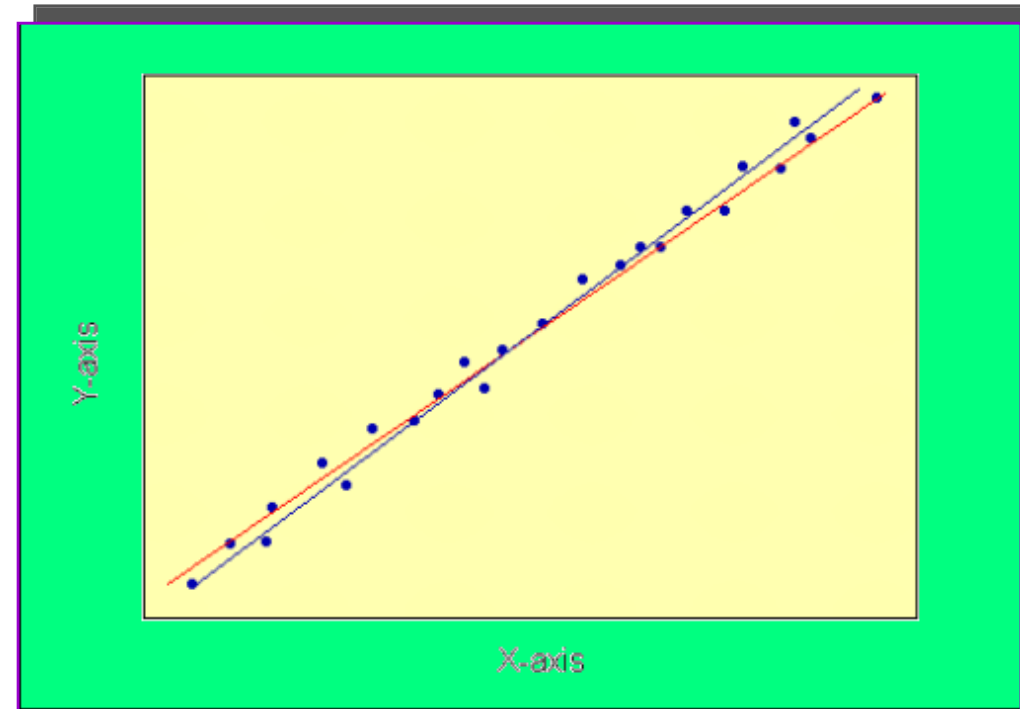


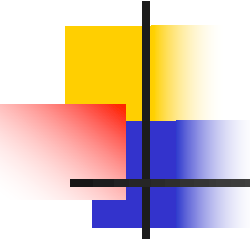
Least-Squares regression line

- In investigating the relationship between two variables, the first thing one should do is to prepare a scatter plot after the data are collected.
- From the plot one can observe any pattern.
- If the correlation coefficient is reasonably large (positive or negative), the next step would be to fit the regression line which best fits or models the data.

Least-Squares regression line

- The scatter plot shows two possible straight lines that may be used to model the data.
- **Question:** Which of these lines best represents the association between the two variables?



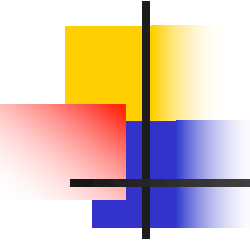


Least-Squares regression line

NOTE: Regression analysis allows us to determine which of the two lines best represents the relationship.

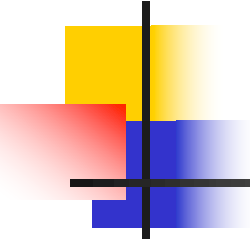
The equation of the linear regression line is usually written as (where *a* is the slope and *b* is the y-intercept)

$$\hat{y} = ax + b$$



Least-Squares regression line

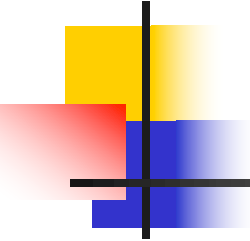
- Least-squares analysis allows us to determine the values for a and b such that the equation of the regression line best represents the relationship between the two variables by minimizing the error sum of squares.
- The regression line is usually called the *line of best fit*.
- We usually refer to this type of regression analysis as **simple regression analysis** since we are dealing with straight-line models involving one independent variable.



Least-Squares regression line

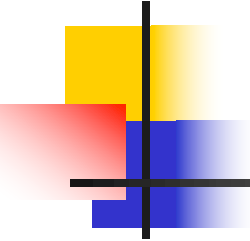
The equations that one can use to compute the values for a and b are:

$$a = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$
$$b = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$



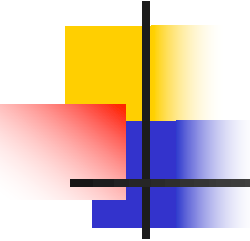
Least-Squares regression line

- Example: The following data relate the high temperature ($^{\circ}\text{F}$) reached on a given day and the number of cans of soft drinks sold from a particular vending machine in front of grocery store. Data were collected for 15 days.
- The data is given on the next slide.



Least-Squares regression line

Temp.	Quantity	Temp.	Quantity	Temp.	Quantity
70	30	98	59	90	53
75	31	72	33	95	56
80	40	75	38	98	62
90	52	75	32	91	51
93	57	80	45	98	58



R: building linear models

```
>t<-read.table("c:/linetemp.txt",header=TRUE)
>model<-lm(temp~quant,t)
>model
```

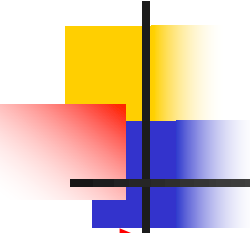
Call:

```
lm(formula = quant ~ temp, data = t)
```

Coefficients:

(Intercept)	temp
-46.171	1.086

$$Y = 1.086 X + 46.171$$



R: building linear models

>summary(model)

Call:

```
lm(formula = quant ~ temp, data = t)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.2489	-1.4179	0.1791	1.6249	4.3232

Coefficients:

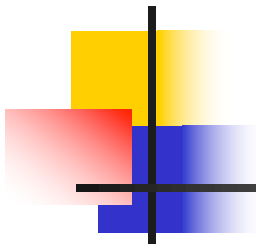
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-46.17065	5.39131	-8.564	1.05e-06 ***
temp	1.08559	0.06275	17.300	2.35e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.43 on 13 degrees of freedom

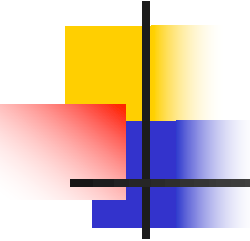
Multiple R-Squared: 0.9584, Adjusted R-squared: 0.9552

F-statistic: 299.3 on 1 and 13 DF, p-value: 2.347e-10



Quick tip

- When using the line of best fit to make predictions, care must be taken to ***use independent values that are within the range of the observed independent variable.***
- Using values outside of the range of observed independent values may lead to incorrect predictions because we do not know how the model is behaving outside this range.
- The model ***reflects the behavior of the association between the two variables only within the range of the observed values.***



Least-Squares regression line

- **Example:** For the previous example, what is the predicted number of soft-drinks sold for a temperature of 85 °F?
- The predicted number of soft-drinks sold is

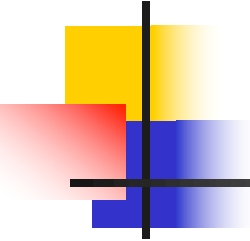
```
d<-data.frame(temp=85)  
> predict(model,d)  
[1] 46.1048
```

The Coefficient of Determination

- The coefficient of determination measures the proportion of the variability in the dependent variable (**y variable**) that is explained by the regression model through the independent variable (**x variable**).
- The coefficient of determination is obtained by squaring the value of the correlation coefficient.

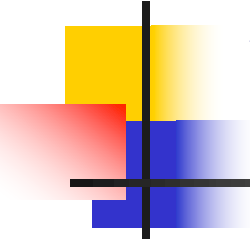
The notation used is r^2 or R^2 .

Note: $0 \leq R^2 \leq 1$ or $0 \leq R^2 \leq 100\%$



The Coefficient of Determination

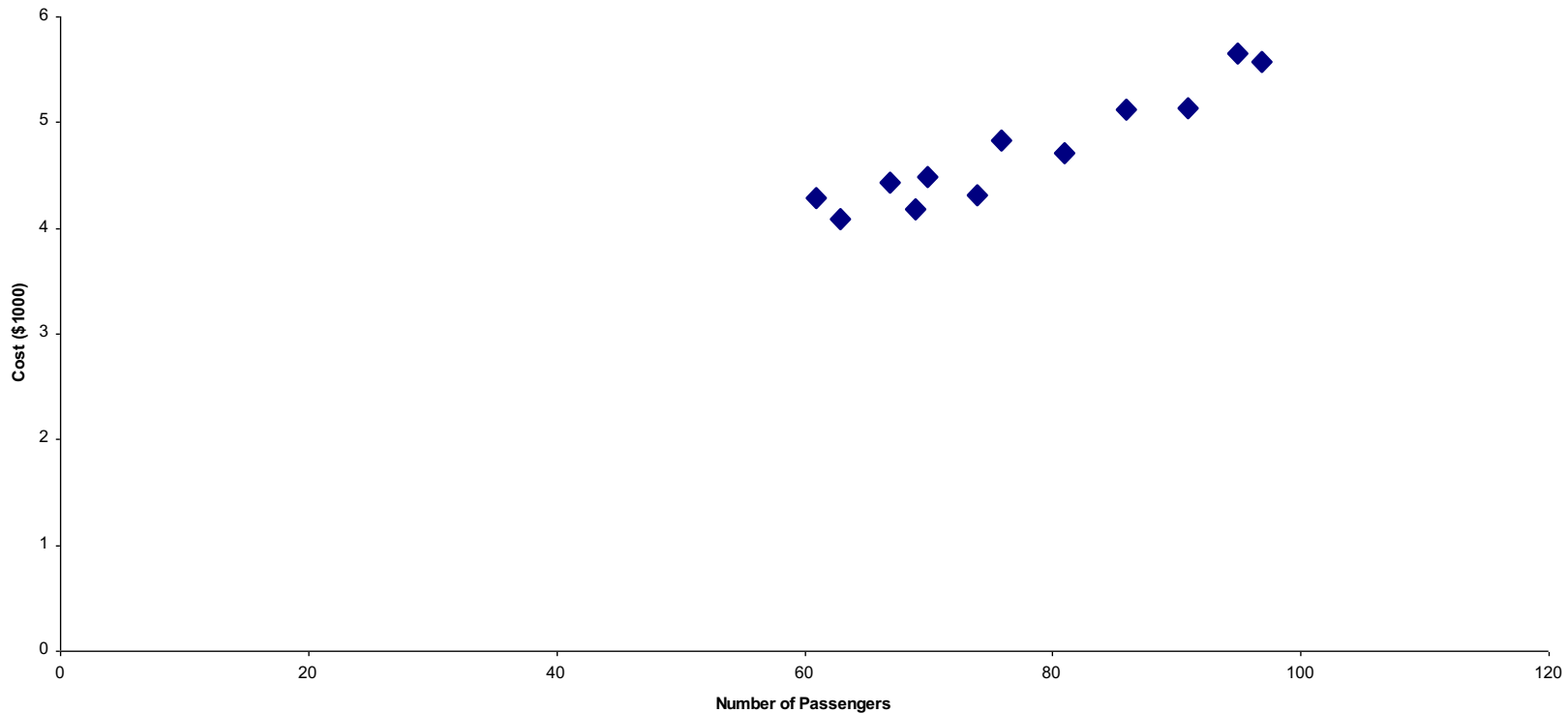
- r^2 or R^2 close to 1 (or 100%) would imply that the model is explaining most of the variation in the dependent variable and may be a very useful model.
- r^2 or R^2 close to 0 (or 0%) would imply that the model is explaining little of the variation in the dependent variable and may not be a very useful model.



Airline Cost Data

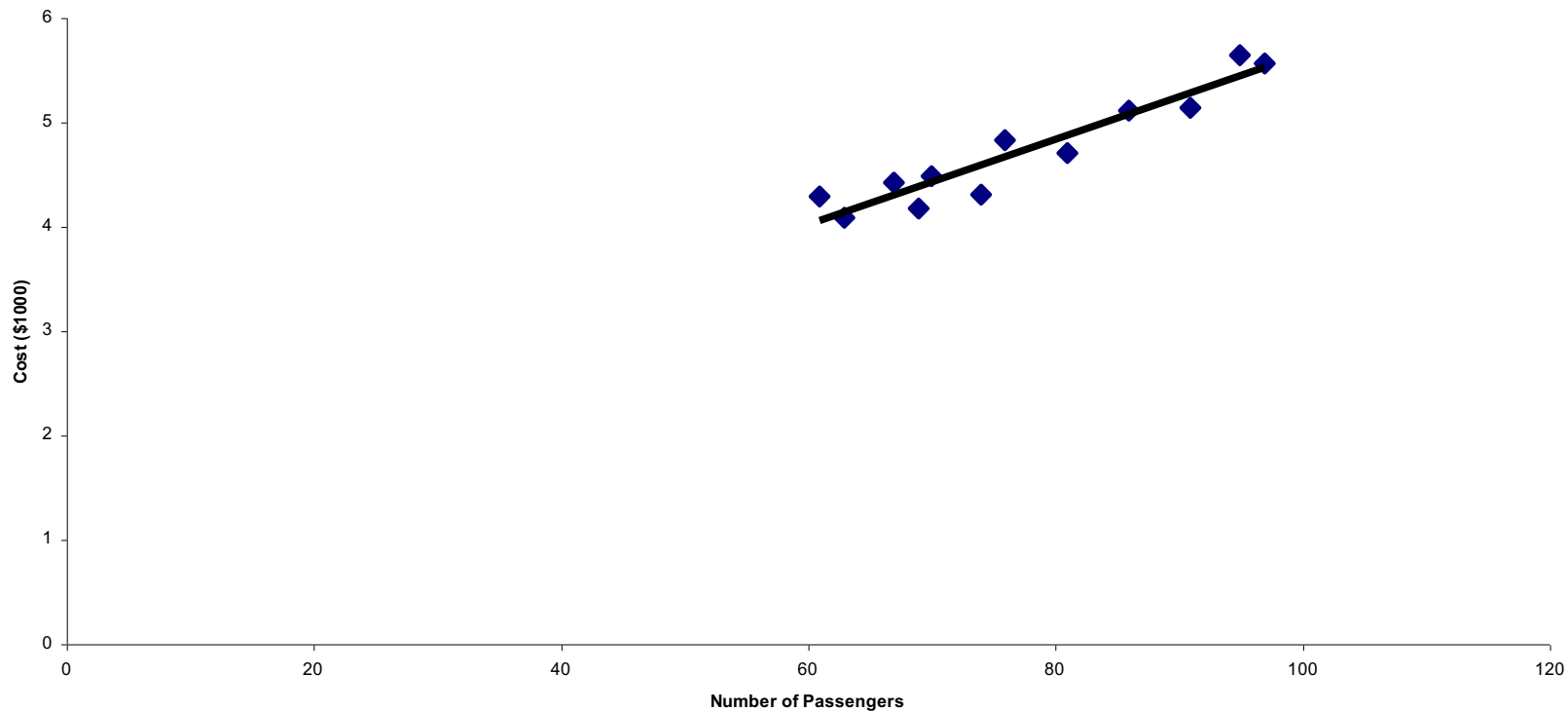
Number of Passengers X	Cost (\$1,000) Y
61	4.280
63	4.080
67	4.420
69	4.170
70	4.480
74	4.300
76	4.820
81	4.700
86	5.110
91	5.130
95	5.640
97	5.560

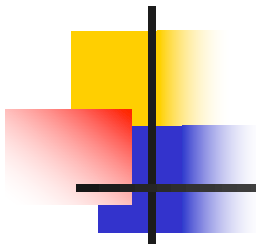
Scatter Plot of Airline Cost Data



Likely to be a linear model...

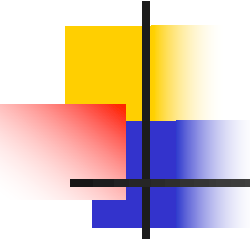
Graph of Regression Line for the Airline Cost Example





Hypotheses

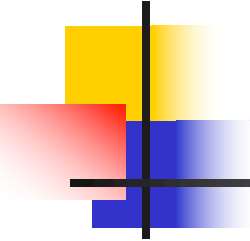
- 1. Normalità:** gli errori per ogni valore di x hanno distribuzione normale; il modello di regressione è robusto rispetto a scostamenti da tale ipotesi: le inferenze su retta e coefficienti non risultano seriamente compromesse da una distribuzione degli errori solo approssimativamente normale.
- 2. Omoschedasticità:** variabilità costante per ciascun valore x (sia per i valori di x piccoli che elevati gli errori devono variare di un medesimo ammontare); tale ipotesi è cruciale: se non soddisfatta si debbono o trasformare opportunamente i dati o ricorrere a metodi di stima diversa (M.Q. ponderati)
- 3. Indipendenza:** per ciascun valore di x ; importante quando i dati sono frutto di osservazioni nel corso del tempo: gli errori prodottisi in un certo periodo potrebbero essere correlati con gli errori del periodo precedente



Residual Analysis: Airline Cost Example

Number of Passengers X	Cost (\$1,000) Y	Predicted Value $\hat{Y}$	Residual $Y - \hat{Y}$
61	4.28	4.053	.227
63	4.08	4.134	-.054
67	4.42	4.297	.123
69	4.17	4.378	-.208
70	4.48	4.419	.061
74	4.30	4.582	-.282
76	4.82	4.663	.157
81	4.70	4.867	-.167
86	5.11	5.070	.040
91	5.13	5.274	-.144
95	5.64	5.436	.204
97	5.56	5.518	.042

$$\sum(Y - \hat{Y}) = -.001$$



Airline Cost: R

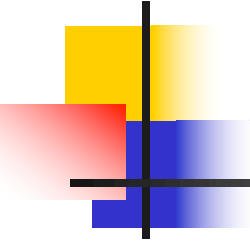
```
t<-read.table("c:/airline.txt",header=TRUE)
```

```
model<-lm(cost~pass,t)
```

```
lm(formula = cost ~ pass, data = t)
```

Coefficients:

(Intercept)	pass
1.5698	0.0407



Airline Cost: R

summary(model)

Call:

```
lm(formula = cost ~ pass, data = t)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.28171	-0.14938	0.04101	0.13162	0.22741

Coefficients:

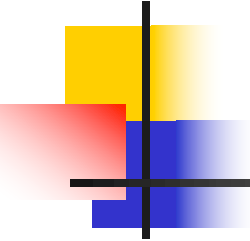
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.569793	0.338083	4.643	0.000917	***
pass	0.040702	0.004312	9.439	2.69e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1772 on 10 degrees of freedom

Multiple R-Squared: 0.8991, Adjusted R-squared: 0.889

F-statistic: 89.09 on 1 and 10 DF, p-value: 2.692e-06



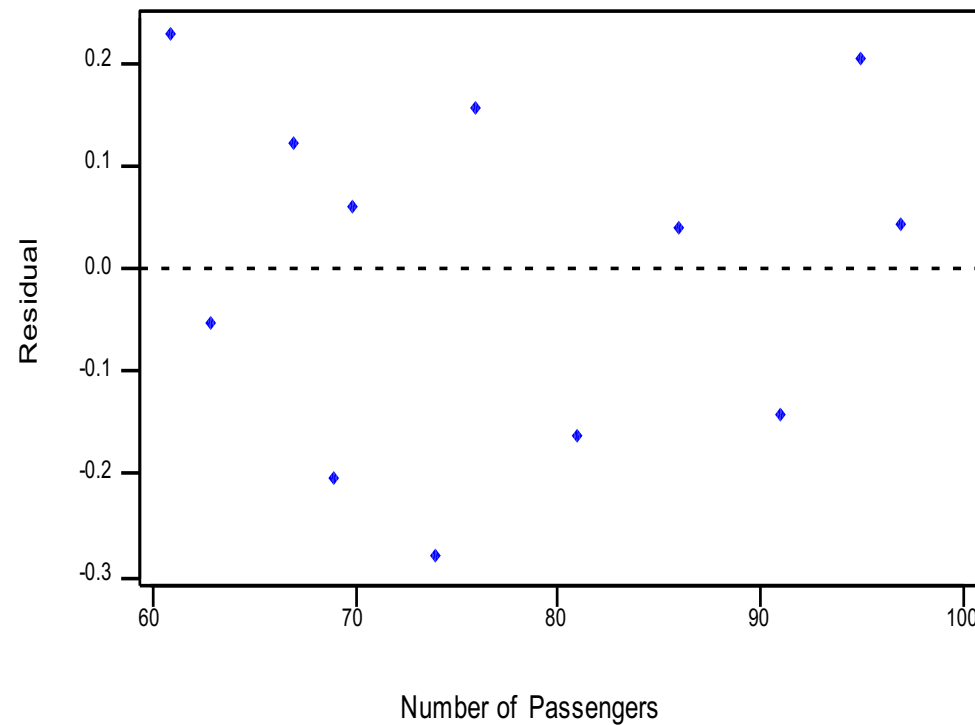
Getting and plotting residuals

- `model$residuals`

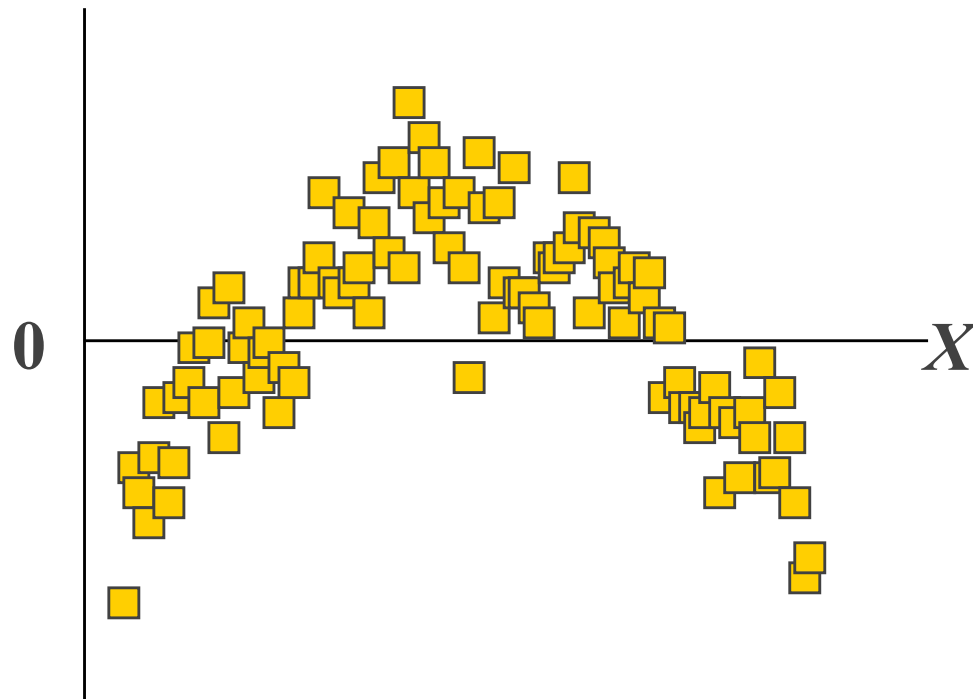
1	2	3	4	5	6
0.22740971	-0.05399349	0.12320012	-0.20820308	0.06109532	-0.28171107
7	8	9	10	11	12
0.15688573	-0.16662226	0.03986975	-0.14363825	0.20355536	0.04215216

- `plot(t$pass,model$residuals)`

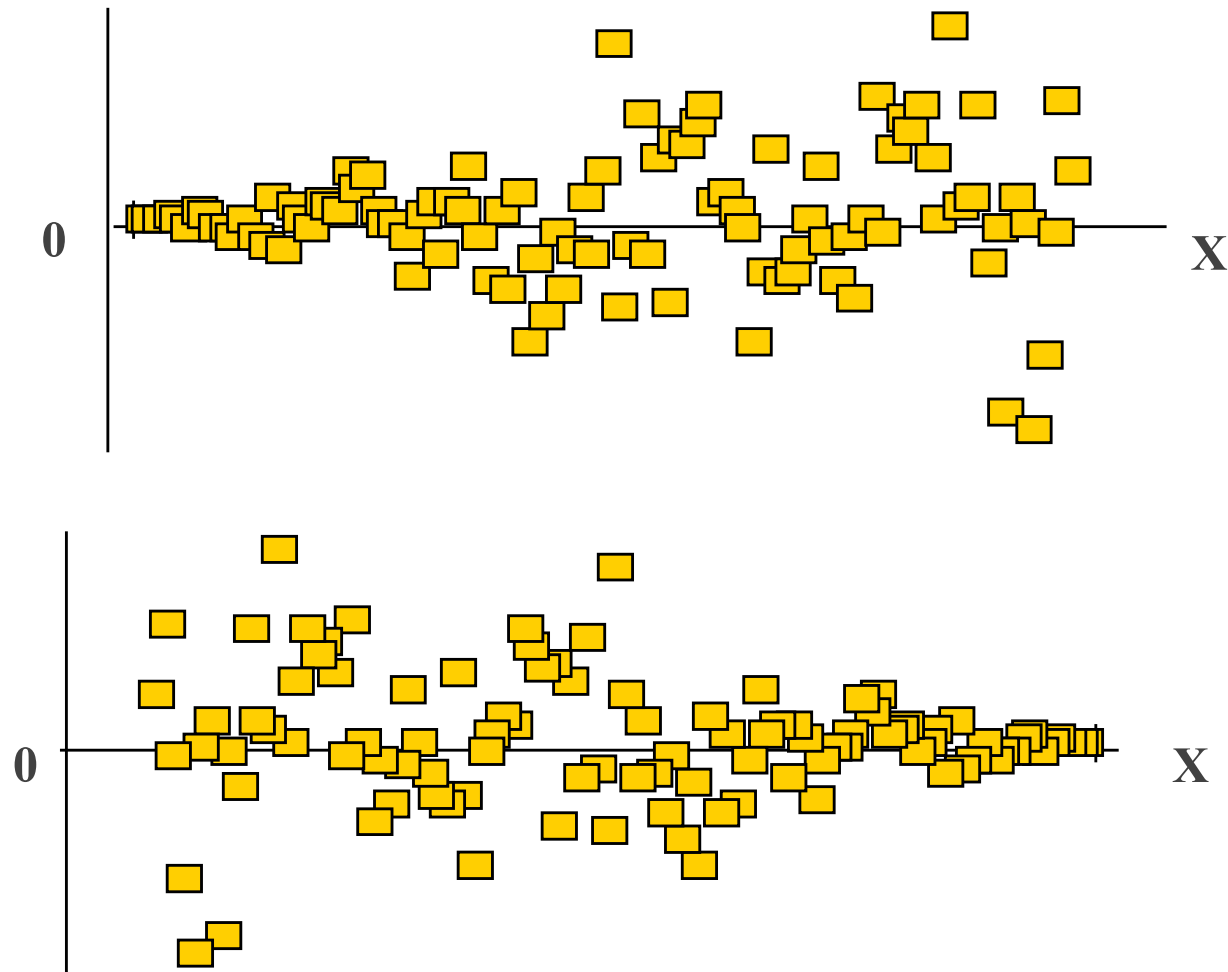
Graph of Residuals for the Airline Cost Example



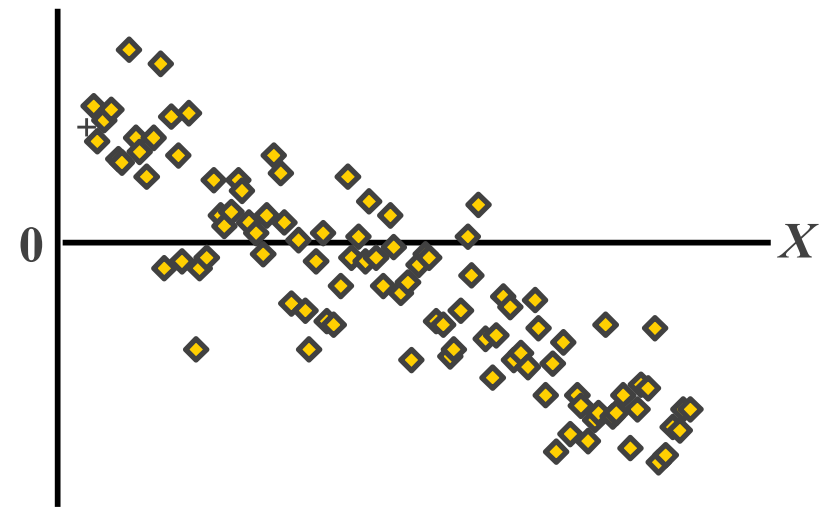
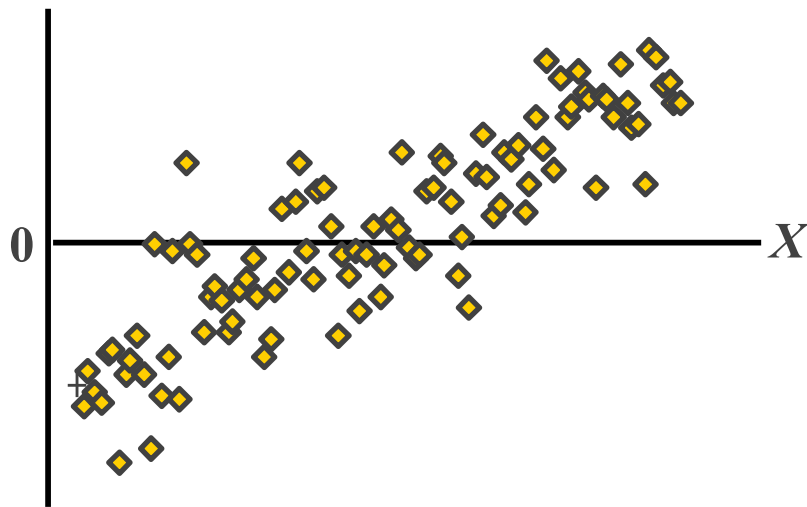
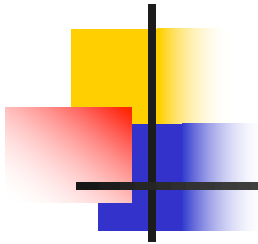
Nonlinear Residual Plot



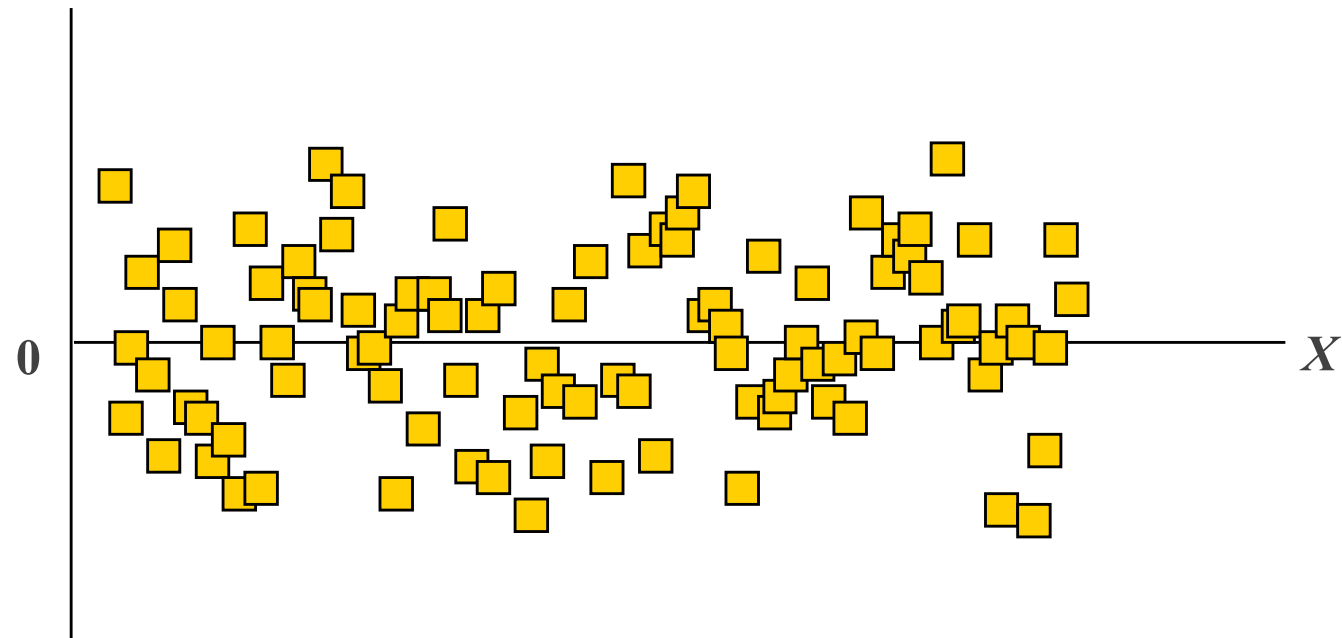
Nonconstant Error Variance

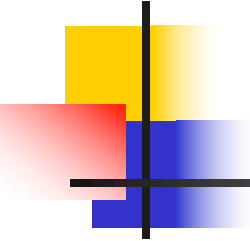


Graphs of Nonindependent Error Terms



Healthy Residual Plot





Verifying the hypotheses

CFP vs EFF

test di linearità

```
sm.regression(Ind, Dep, h=300,model="linear",test=TRUE, xlab="CFP", ylab="EFF")  
text(Ind, Dep)
```

perform Breusch-Pagan test per l'omoschedasticità

bptest(res)

```
bptest(Dep ~ Ind)
```

linear regression

```
fm <- lm(Dep ~ Ind)
```

```
summary(fm)
```

normalità dei residui ottenuti con il modello

```
shapiro.test(resid(fm))
```

dw<-dwtest(fm,data=dati) ## test di Durbin-Watson

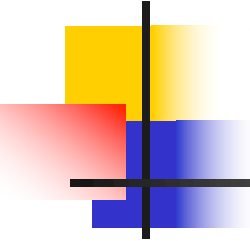
dw

The Durbin-Watson statistic is used to test for the presence of serial correlation

among the residuals. The value of the Durbin-Watson statistic ranges from 0 to 4.

As a general rule of thumb, the residuals are not correlated if the Durbin-Watson

statistic is approximately 2, and an acceptable range is 1.50 - 2.50.

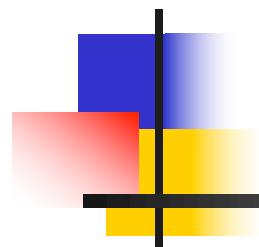


Verifying the hypotheses

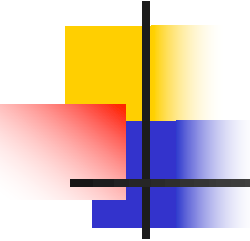
```
### analisi degli outliers  
# distanza di cook  
cooks.distance(fm)
```

```
## plottando i residui
```

```
plot(fitted(fm), resid(fm),  
  xlab="Fitted values",  
  ylab="Residuals",  
  main="Scatter plot of Residuals"  
)  
text(fitted(fm), resid(fm))
```



Multiple Regression Analysis



Regression Models

Probabilistic Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

Y = the value of the dependent (response) variable

β_0 = the regression constant

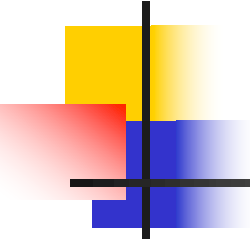
β_1 = the partial regression coefficient of independent variable 1

β_2 = the partial regression coefficient of independent variable 2

β_k = the partial regression coefficient of independent variable k

k = the number of independent variables

ε = the error of prediction



Estimated Regression Model

$$\hat{Y} = \mathbf{b}_0 + \mathbf{b}_1 \mathbf{X}_1 + \mathbf{b}_2 \mathbf{X}_2 + \mathbf{b}_3 \mathbf{X}_3 + \cdots + \mathbf{b}_k \mathbf{X}_k$$

where : $\hat{Y}$ = predicted value of Y

$\mathbf{b}_0$ = estimate of regression constant

$\mathbf{b}_1$ = estimate of regression coefficient 1

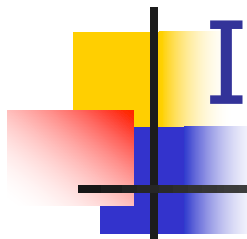
$\mathbf{b}_2$ = estimate of regression coefficient 2

$\mathbf{b}_3$ = estimate of regression coefficient 3

$\mathbf{b}_k$ = estimate of regression coefficient k

k = number of independent variables

Multiple Regression Model with Two Independent Variables (First-Order)


$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

**Population
Model**

where : β_0 = the regression constant

β_1 = the partial regression coefficient for independent variable 1

β_2 = the partial regression coefficient for independent variable 2

ε = the error of prediction

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$$

**Estimated
Model**

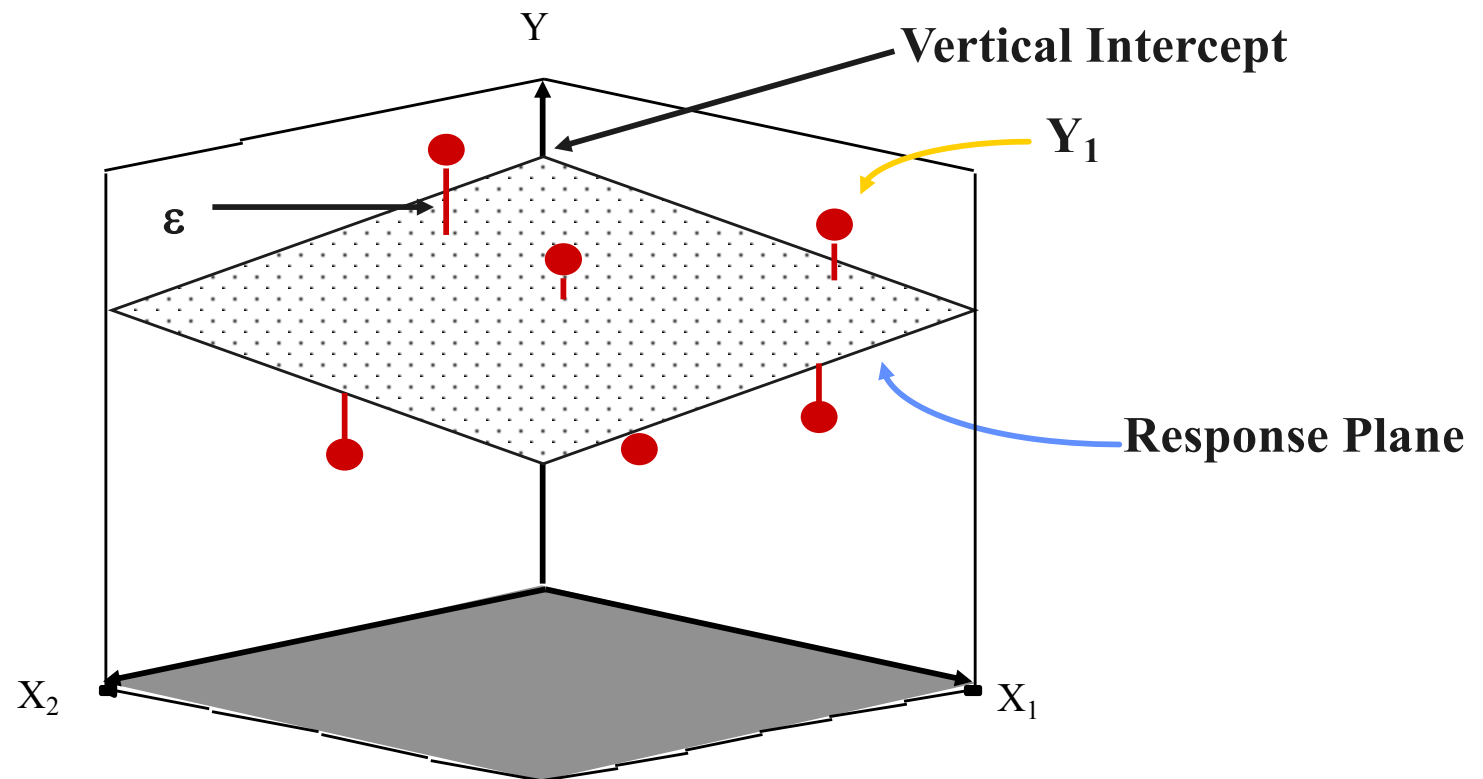
where : $\hat{Y}$ = predicted value of Y

b_0 = estimate of regression constant

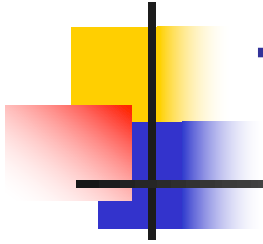
b_1 = estimate of regression coefficient 1

b_2 = estimate of regression coefficient 2

Response Plane for First-Order Two-Predictor Multiple Regression Model



Least Squares Equations for $k = 2$



$$b_0 n + b_1 \sum X_1 + b_2 \sum X_2 = \sum Y$$

$$b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2 = \sum X_1 Y$$

$$b_0 \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2 = \sum X_2 Y$$

Real Estate Data



	Market Price (\$1,000)	Square Feet	Age (Years)		Market Price (\$1,000)	Square Feet	Age (Years)
Observation	Y	X ₁	X ₂	Observation	Y	X ₁	X ₂
1	63.0	1,605	35	13	79.7	2,121	14
2	65.1	2,489	45	14	84.5	2,485	9
3	69.9	1,553	20	15	96.0	2,300	19
4	76.8	2,404	32	16	109.5	2,714	4
5	73.9	1,884	25	17	102.5	2,463	5
6	77.9	1,558	14	18	121.0	3,076	7
7	74.9	1,748	8	19	104.9	3,048	3
8	78.0	3,105	10	20	128.0	3,267	6
9	79.0	1,682	28	21	129.0	3,069	10
10	63.4	2,470	30	22	117.9	4,765	11
11	79.5	1,820	2	23	140.0	4,540	8
12	83.9	2,143	6				



```
t<-read.table("c:/real-estate.txt",header=TRUE)
model<-lm(Price~Feet+Age,t)
```

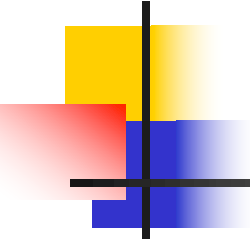
Call:

```
lm(formula = Price ~ Feet + Age, data = t)
```

Coefficients:

(Intercept)	Feet	Age
59.08481	0.01732	-0.77132

Price=59.08481 + 0.01732*Feet -0.77132*Age



R - summary

`summary(model)`

Call:

```
lm(formula = Price ~ Feet + Age, data = t)
```

Residuals:

Min	1Q	Median	3Q	Max
-27.1593	-7.9873	0.7531	7.4628	24.4643

Coefficients:

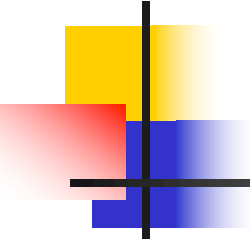
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	59.084812	10.453872	5.652	1.57e-05 ***
Feet	0.017323	0.003286	5.272	3.69e-05 ***
Age	-0.771322	0.238175	-3.238	0.00412 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.49 on 20 degrees of freedom

Multiple R-Squared: 0.735, Adjusted R-squared: 0.7085

F-statistic: 27.73 on 2 and 20 DF, p-value: 1.708e-06



Predicting the Price of Home

```
> d<-data.frame(Feet=2500,Age=12)
```

```
d
```

```
  Feet Age
```

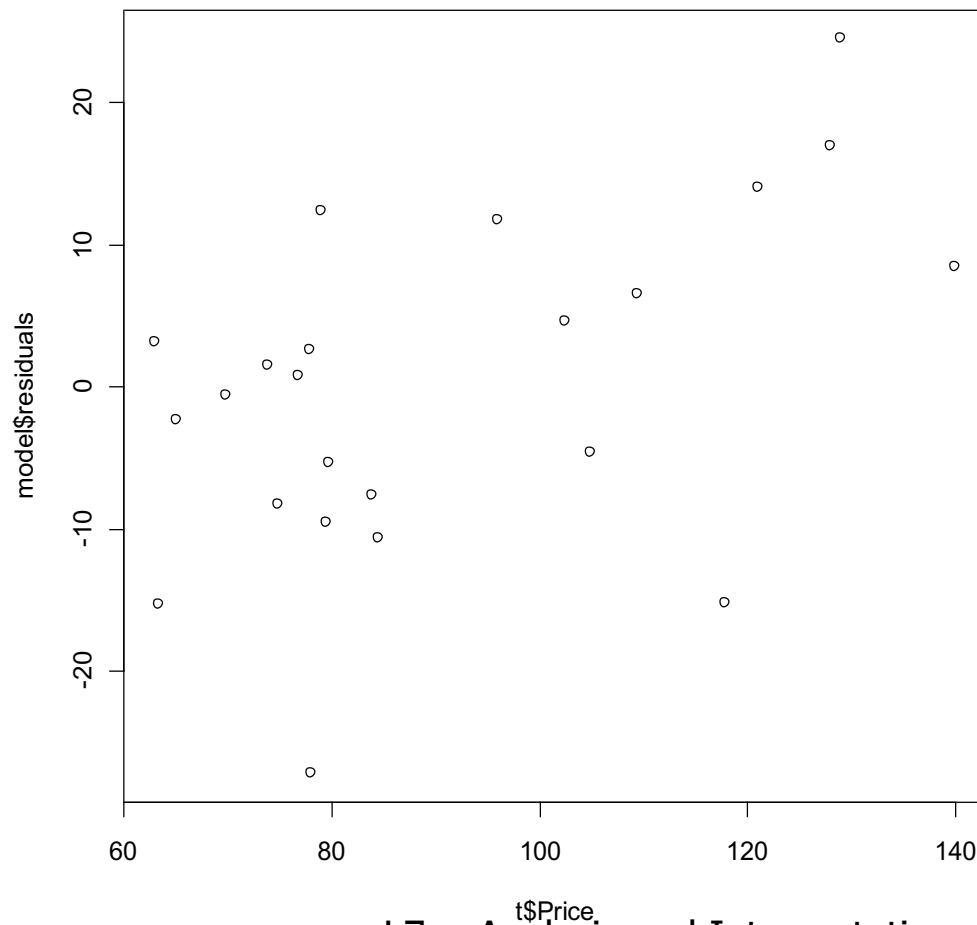
```
1 2500  12
```

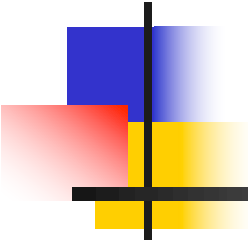
```
> predict(model,d)
```

```
[1] 93.1363
```

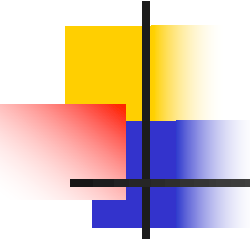
Residuals

```
plot(t$Price,model$residuals)
```





General Regression Models



General Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

Y = the value of the dependent (response) variable

β_0 = the regression constant

β_1 = the partial regression coefficient of independent variable 1

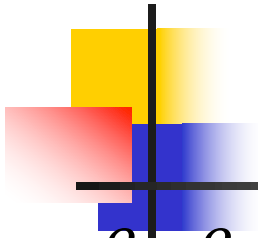
β_2 = the partial regression coefficient of independent variable 2

β_k = the partial regression coefficient of independent variable k

k = the number of independent variables

ε = the error of prediction

Non Linear Models: Mathematical Transformation



$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

First-order with Two Independent Variables

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \varepsilon$$

Second-order with One Independent Variable

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

**Second-order with an
Interaction Term**

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 + \varepsilon$$

**Second-order with
Two Independent
Variables**

Nonlinear Regression Models: Model Transformation

$$Y = \beta_0 \beta_1^X \varepsilon$$

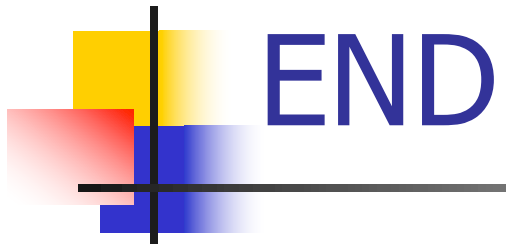
$$\log Y = \log \beta_0 + X \log \beta_1$$

$$\hat{Y}' = b_0' + b_1' X$$

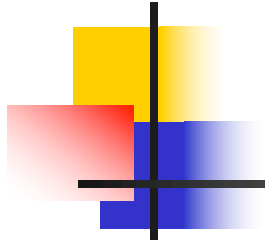
$$\text{where: } \hat{Y}' = \log \hat{Y}$$

$$b_0' = \log b_0$$

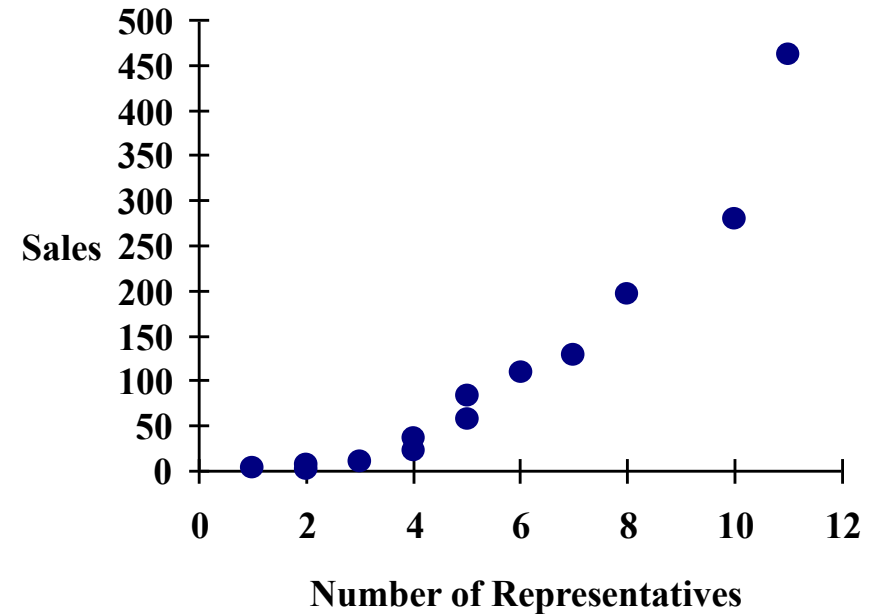
$$b_1' = \log b_1$$

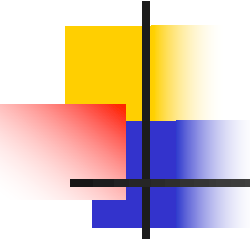


Sales Data and Scatter Plot for 13 Manufacturing Companies



Manufacturer	Sales (\$1,000,000)	Number of Manufacturing Representatives
1	2.1	2
2	3.6	1
3	6.2	2
4	10.4	3
5	22.8	4
6	35.6	4
7	57.1	5
8	83.5	5
9	109.4	6
10	128.6	7
11	196.8	8
12	280.0	10
13	462.3	11





R: simple linear model

```
>t<-read.table("c:/sales.txt",header=TRUE)
>model<-lm(sales~repr,t)
```

Call:

```
lm(formula = sales ~ repr, data = t)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.55	-29.73	-21.48	27.08	118.04

Coefficients:

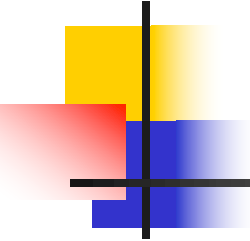
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-107.029	28.737	-3.724	0.00336	**
repr	41.026	4.779	8.584	3.32e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 51.1 on 11 degrees of freedom

Multiple R-Squared: 0.8701, Adjusted R-squared: 0.8583

F-statistic: 73.69 on 1 and 11 DF, p-value: 3.322e-06

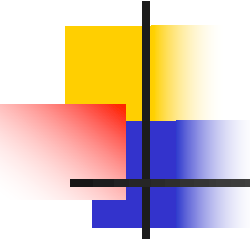


R: simple linear model

```
summary(aov(model))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
repr	1	192395	192395	73.685	3.322e-06 ***
Residuals	11	28721	2611		

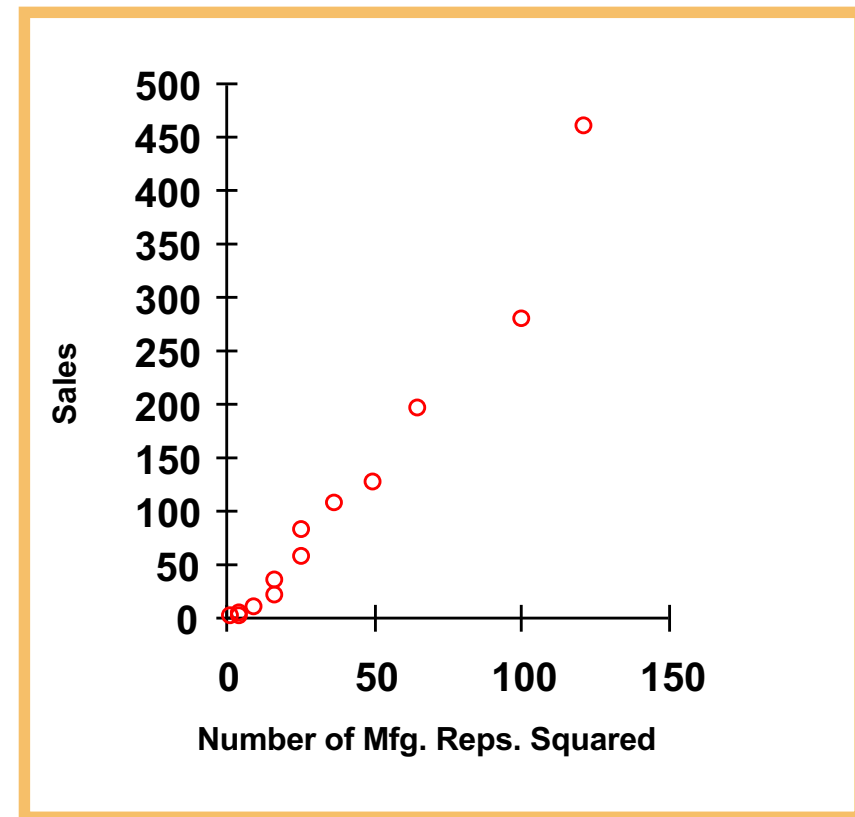
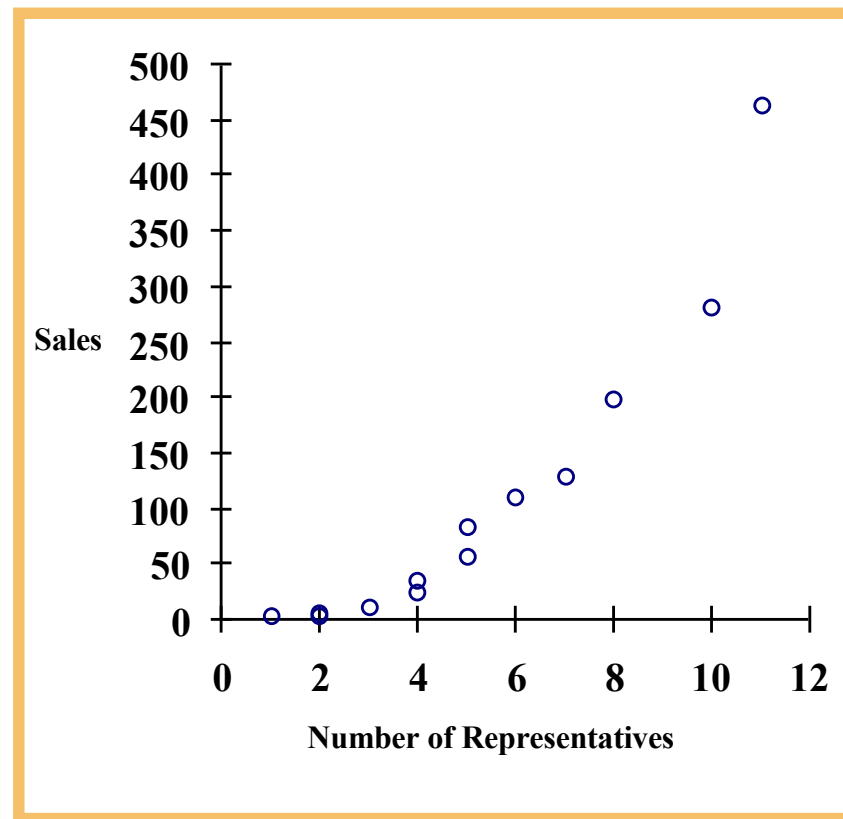
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

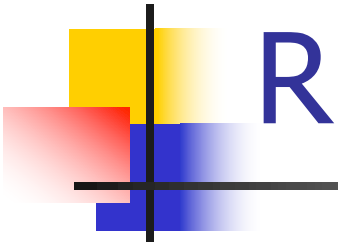


Manufacturing Data with Newly Created Variable

Manufacturer	Sales (\$1,000,000)	Number of Mgfr Reps X_1	(No. Mgfr Reps)² $X_2 = (X_1)^2$
1	2.1	2	4
2	3.6	1	1
3	6.2	2	4
4	10.4	3	9
5	22.8	4	16
6	35.6	4	16
7	57.1	5	25
8	83.5	5	25
9	109.4	6	36
10	128.6	7	49
11	196.8	8	64
12	280.0	10	100
13	462.3	11	121

Scatter Plots Using Original and Transformed Data





- #original boxplot
- `plot(t$repr,t$sales)`
- `r2<-t$repr^2`
- #transformed boxplot
- `plot(r2,t$sales)`

R:

Quadratic Model to Predict Sales

```
mod2<-lm(t$sales~r2)
summary(mod2)
```

Call:

```
lm(formula = t$sales ~ r2)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.708	-11.312	-1.217	11.510	56.870

Coefficients:

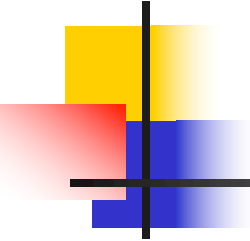
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-19.3526	10.3204	-1.875	0.0875
r2	3.5106	0.2009	17.475	2.26e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 26.44 on 11 degrees of freedom

Multiple R-Squared: 0.9652, Adjusted R-squared: 0.9621

F-statistic: 305.4 on 1 and 11 DF, p-value: 2.261e-09



R:

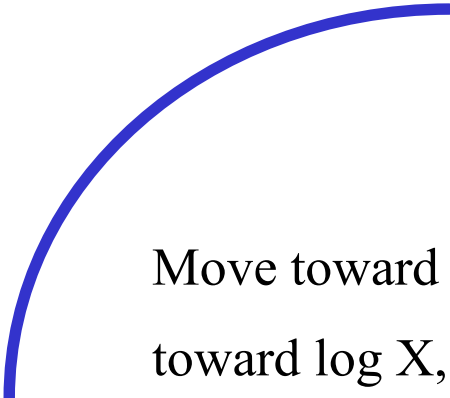
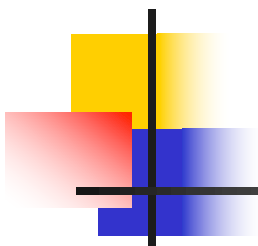
Quadratic Model to Predict Sales

```
summary(aov(mod2))
```

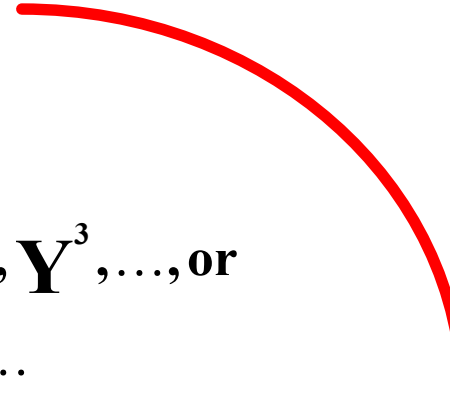
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
r2	1	213429	213429	305.39	2.261e-09 ***
Residuals	11	7688	699		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

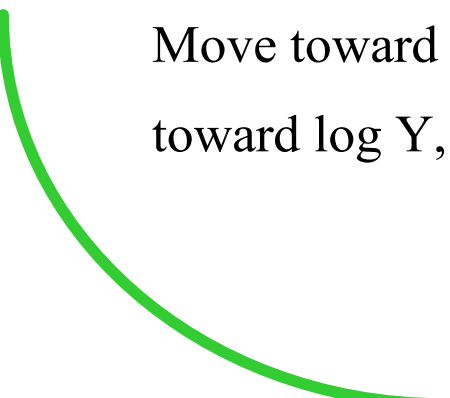
Tukey's Four Quadrant Approach



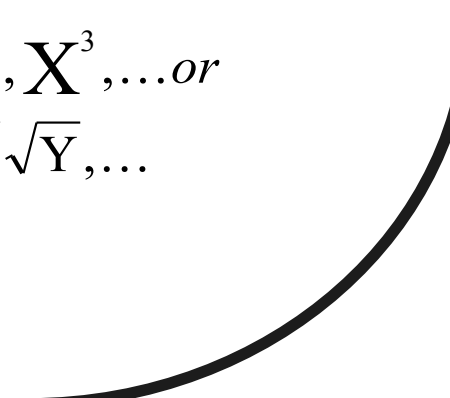
Move toward $Y^2, Y^3, \dots$, or
toward $\log X, -1/\sqrt{X}, \dots$



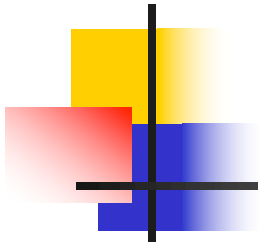
**Move toward $Y^2, Y^3, \dots$, or
toward $X^2, X^3, \dots$**



Move toward $\log X, -1/\sqrt{X}, \dots$, or
toward $\log Y, -1/\sqrt{Y}, \dots$

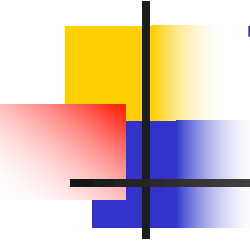


Move toward $X^2, X^3, \dots$, or
toward $\log Y, -1/\sqrt{Y}, \dots$



Prices of Three Stocks over a 15- Month Period

Stock 1	Stock 2	Stock 3
41	36	35
39	36	35
38	38	32
45	51	41
41	52	39
43	55	55
47	57	52
49	58	54
41	62	65
35	70	77
36	72	75
39	74	74
33	83	81
28	101	92
31	107	91



Regression Models for the Three Stocks

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where: Y = price of stock 1

X_1 = price of stock 2

X_2 = price of stock 3

**First-order with
Two Independent Variables**

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

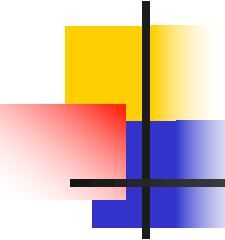
where: Y = price of stock 1

X_1 = price of stock 2

X_2 = price of stock 3

$X_3 = X_1 X_2$

**Second-order with an
Interaction Term**



Regression for Three Stocks: First-order, Two Independent Variables

```
>t<-read.table("c:/stock.txt",header=TRUE)
>model<-lm(s1~s2+s3,t)
>summary(model)
```

Call:

```
lm(formula = s1 ~ s2 + s3, data = t)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0691	-2.6708	-0.9078	2.3842	8.8677

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	50.85548	3.79099	13.415	1.38e-08 ***
s2	-0.11900	0.19308	-0.616	0.549
s3	-0.07076	0.19898	-0.356	0.728

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.57 on 12 degrees of freedom

Multiple R-Squared: 0.4723, Adjusted R-squared: 0.3843

F-statistic: 5.369 on 2 and 12 DF, p-value: 0.02160

Regression for Three Stocks: Second-order With an Interaction Term

```
>model<-lm(s1~s2*s3,t)
> summary(model)
```

Call:

```
lm(formula = s1 ~ s2 * s3, data = t)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-5.0924	-1.8280	0.1816	1.3908	5.3510

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.046177	9.312400	1.294	0.22232
s2	0.878778	0.261873	3.356	0.00641 **
s3	0.220493	0.143522	1.536	0.15271
s2:s3	-0.009985	0.002314	-4.315	0.00123 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.909 on 11 degrees of freedom

Multiple R-Squared: 0.804, Adjusted R-squared: 0.7505

F-statistic: 15.04 on 3 and 11 DF, p-value: 0.00033

L7 - Analysis and Interpretation

Nonlinear Regression Models: Model Transformation

$$Y = \beta_0 \beta_1^X \varepsilon$$

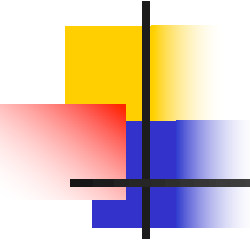
$$\log Y = \log \beta_0 + X \log \beta_1$$

$$\hat{Y}' = b_0' + b_1' X$$

$$\text{where: } \hat{Y}' = \log \hat{Y}$$

$$b_0' = \log b_0$$

$$b_1' = \log b_1$$



Data Set for Model Transformation Example

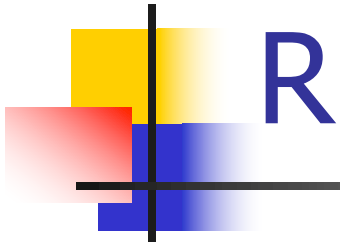
ORIGINAL DATA

Company	Y	X
1	2580	1.2
2	11942	2.6
3	9845	2.2
4	27800	3.2
5	18926	2.9
6	4800	1.5
7	14550	2.7

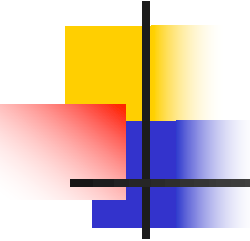
TRANSFORMED DATA

Company	LOG Y	X
1	3.41162	1.2
2	4.077077	2.6
3	3.993216	2.2
4	4.444045	3.2
5	4.277059	2.9
6	3.681241	1.5
7	4.162863	2.7

Y = Sales (\$ million/year) X = Advertising (\$ million/year)



- `t<-`
`read.table("c:/company.txt",header=TRUE)`
- `ly<-log10(t$Y)`
- `model<-lm(ly~X,t)`



Regression Output for Model Transformation Example

summary(model)

Call:

```
lm(formula = ly ~ X, data = t)
```

Residuals:

1	2	3	4	5	6	7
-0.058897	-0.058617	0.047572	0.023274	-0.001174	0.068186	-0.020344

Coefficients:

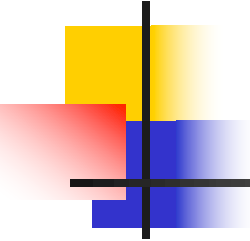
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.90036	0.07291	39.78	1.89e-07	***
X	0.47513	0.03004	15.82	1.84e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05436 on 5 degrees of freedom

Multiple R-Squared: 0.9804, Adjusted R-squared: 0.9765

F-statistic: 250.1 on 1 and 5 DF, p-value: 1.838e-05



Prediction with the Transformed Model

- By doing a prediction, we obtain $\log Y$
 - `d<-data.frame(X=2)`
 - `10^predict(model,d)`
 - `[1] 7089.545`